

PETERSBURG, AK, USA

WMO: 703860

Lat: 56.806N

Lon: 132.937W

Elev: 107

StdP: 14.64

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25329

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.6	(c) 12.4	(d) 0.9	(e) 5.8	(f) 13.7	(g) 6.7	(h) 7.8	(i) 16.4	(j) 17.4	(k) 40.5	(l) 15.7	(m) 39.9	(n) 2.9	(o) 250	(p) 0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	69.9	58.3	66.0	57.0	63.5	55.5	59.5	66.0	58.2	64.2	56.9	62.3	5.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.6	68.5	61.2	55.1	64.8	59.7	54.3	63.1	58.9	26.0	66.4	25.1	63.9	24.3	62.1	64.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.7	(b) 12.1	(c) 10.4	(e) 3.7	(f) 76.5	(g) 3.9	(h) 4.0	(i) 0.9	(j) 79.4	(k) -1.4	(l) 81.7	(m) -3.6	(n) 83.9	(o) -6.4	(p) 86.8		
(a) 14.7	(b) 12.1	(c) 10.4	(e) 3.5	(f) 62.0	(g) 4.0	(h) 1.5	(i) 0.6	(j) 63.1	(k) -1.7	(l) 64.0	(m) -3.9	(n) 64.8	(o) -6.8	(p) 66.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	31.6	32.7	34.7	40.8	47.8	53.3	56.2	55.6	50.5	43.4	36.0	32.2		
	DBStd	10.75	8.94	7.06	6.15	4.01	4.84	4.09	3.24	3.23	3.40	4.52	6.82	8.13		
	HDD50	3134	570	485	476	278	101	9	1	1	32	208	420	553		
	HDD65	8048	1035	905	941	727	532	352	274	291	434	669	870	1018		
	CDD50	562	0	0	0	1	34	108	192	175	48	4	0	0		
	CDD65	1	0	0	0	0	0	1	0	0	0	0	0	0		
	CDH74	14	0	0	0	0	0	9	3	2	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	3.6	3.4	3.3	3.9	4.2	4.2	3.8	3.2	2.9	3.0	3.5	3.6	3.6		
	PrecAvg	100.8	9.0	7.3	7.2	5.3	4.6	3.9	4.9	6.3	12.5	13.6	11.6	10.1		
(15) Precipitation	PrecMax	128.2	18.6	14.7	14.9	9.7	13.1	10.9	8.8	15.9	19.9	22.1	21.9	22.2		
	PrecMin	56.2	0.9	2.2	0.9	0.4	0.0	0.6	1.6	2.0	6.8	1.3	2.8	1.7		
	PrecStd	16.1	4.1	3.3	3.3	2.4	3.0	2.2	2.0	3.3	4.1	4.6	4.8	5.4		
	PrecStd	16.1	4.1	3.3	3.3	2.4	3.0	2.2	2.0	3.3	4.1	4.6	4.8	5.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	50.4	52.2	60.5	69.0	74.6	72.7	72.2	63.7	57.5	54.4	51.9		
		MCWB	42.2	41.8	44.2	47.4	54.7	58.2	60.0	59.0	54.7	50.5	44.9	41.9		
	2%	DB	46.4	46.4	46.8	54.3	63.6	68.2	69.6	68.0	60.9	54.5	48.5	47.7		
		MCWB	42.0	40.6	41.3	45.0	52.6	56.4	59.2	58.3	53.6	50.1	44.9	43.2		
	5%	DB	44.9	45.0	45.0	51.6	60.6	63.9	65.5	64.3	57.3	52.1	46.1	45.3		
		MCWB	40.9	40.1	40.2	43.8	51.1	55.0	57.9	56.9	52.2	48.5	42.8	42.0		
	10%	DB	42.6	42.7	42.9	48.0	56.6	61.2	63.1	62.7	55.4	49.8	44.8	42.9		
		MCWB	39.4	38.8	38.6	42.3	49.2	53.6	56.2	56.2	51.8	47.0	41.3	39.6		
	0.4%	WB	45.3	43.9	45.8	49.0	56.1	59.8	61.6	61.5	58.1	53.4	47.5	46.2		
		MCDB	47.2	47.3	50.1	56.5	66.1	70.6	70.4	69.3	62.5	55.8	50.4	48.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.8	41.5	42.4	46.3	53.5	57.6	59.6	59.4	55.3	50.8	45.7	44.1		
		MCDB	46.0	45.7	46.3	51.9	61.5	65.8	66.3	65.0	58.8	53.5	48.1	46.5		
	5%	WB	41.0	40.3	40.8	44.5	51.6	56.0	58.3	58.2	53.9	48.7	43.5	41.5		
		MCDB	44.8	44.5	44.5	48.9	57.9	63.0	64.1	63.4	56.1	51.1	46.2	45.3		
	10%	WB	39.3	38.9	39.2	43.1	50.3	54.3	56.9	56.8	52.9	46.9	41.5	39.7		
		MCDB	42.6	42.4	42.5	47.4	55.8	60.1	62.1	61.5	55.1	48.7	44.7	43.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	9.5	10.7	12.8	13.2	11.6	10.6	10.9	10.0	8.7	7.7	7.5		
		MCDBR	9.8	11.4	12.9	18.5	20.1	18.5	17.9	17.0	12.9	11.8	9.7	9.9		
		MCWBR	6.4	6.7	8.5	10.9	10.8	9.9	9.7	9.6	8.3	7.8	6.2	6.7		
		MCDBR	9.3	10.3	11.8	14.9	16.2	16.1	14.2	14.1	10.2	10.0	8.9	9.7		
	5% WB	MCWBR	6.4	6.6	8.1	9.7	9.3	9.0	8.3	8.6	7.6	7.1	6.1	6.9		
		MCWBR	6.4	6.6	8.1	9.7	9.3	9.0	8.3	8.6	7.6	7.1	6.1	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.305	0.332	0.379	0.406	0.410	0.414	0.373	0.362	0.327	0.294	0.273		
	taud		2.220	2.297	2.249	2.140	2.134	2.218	2.263	2.421	2.405	2.390	2.217	2.140		
	Ebn at Noon		188	238	260	263	263	263	259	265	249	227	180	154		
	Edh at Noon		17	23	32	41	44	42	39	31	27	21	16	14		
(44) All-Sky Solar Radiation	RadAvg		149	403	772	1250	1572	1558	1355	1135	743	412	187	108		
	RadStd		16	66	94	108	154	137	161	130	110	56	25	15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	+1.80	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page